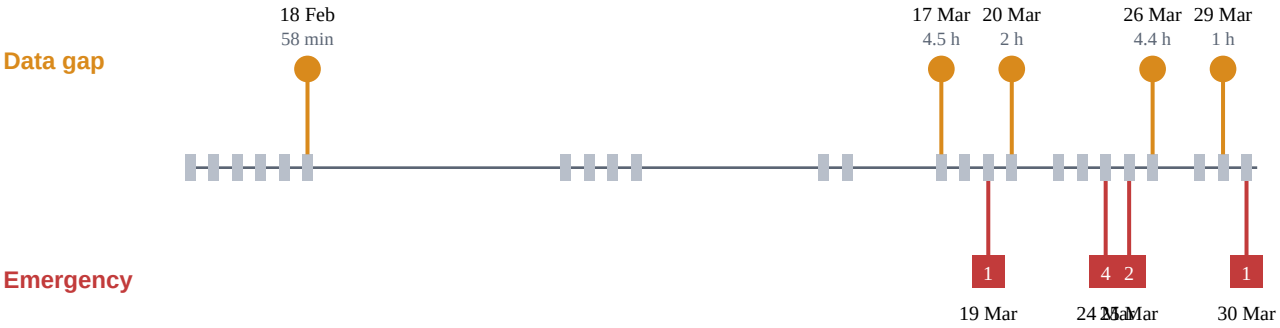
The missing data was not caused by trips or emergency stops

None of the five data gaps falls on a day with an emergency stop or purge. On every one of those days, the logger kept recording straight through the stop.

Why it matters. A gap does not hide a safety event. Two gaps do hide a moment when the plant switched off while hot, and the cause is not in the data. Those two need the operator's event log (page 3).



Grey ticks = days with a log file (13 Feb to 30 Mar 2026). Red number = emergency events that day.

The orange gap days and the red emergency days never meet. The 8 emergency events (6 stops, 2 purges) all happened on 19, 24, 25 and 30 March, and those four days have no gap. 22 March has an empty (0-byte) file.

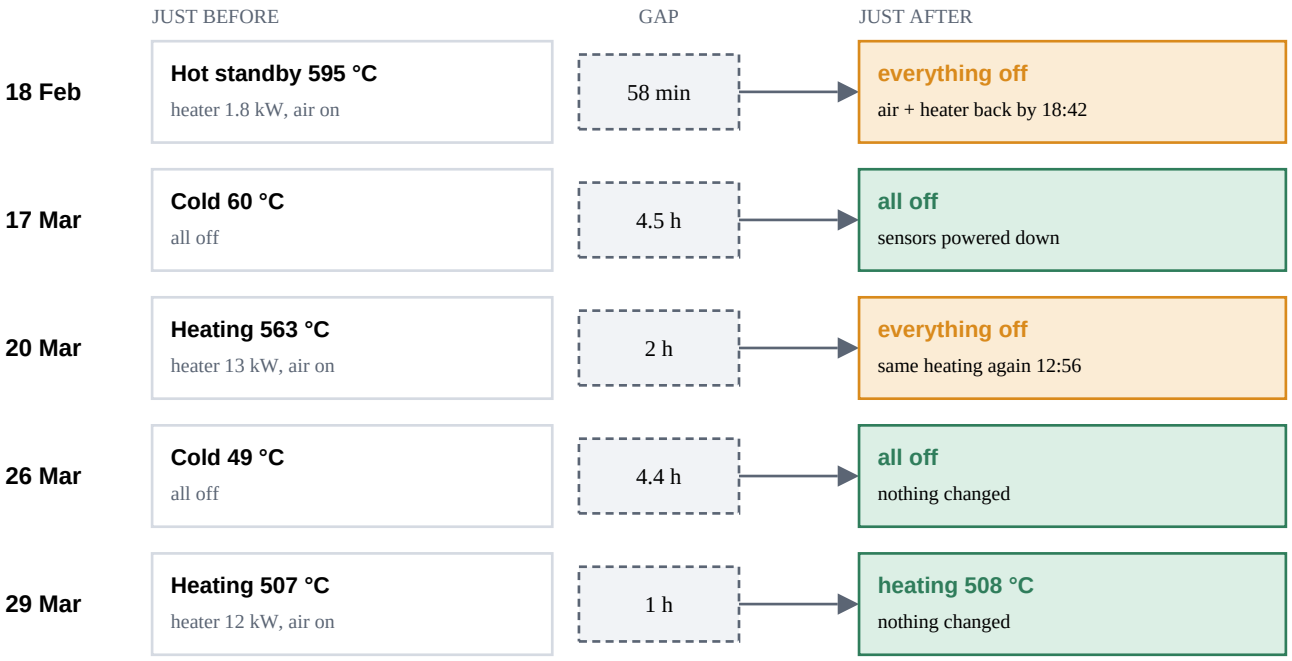
Where the gaps come from

Day	Missing	Plant at the time	Most likely cause
18 Feb	17:54-18:51	Hot, 595 °C, standby	Logger outage, plant found off
17 Mar	10:27-14:55	Cold, 60 °C, all off	Instruments switched off
20 Mar	10:53-12:56	Hot, 563 °C, heating	Logger restart, plant found off
26 Mar	13:03-17:25	Cold, 49 °C, all off	Logger not recording
29 Mar	02:00-03:00	Heating, 507 °C	Clock change (summer time)

If you remember one thing: the gaps are logger problems, not trips. No trip limit was even close when any gap started.

Two gaps hide a stop. Three hide nothing.

For each gap we looked at the last second of good data before it and the first second after it. If the plant looks the same on both sides, nothing happened inside the gap. If it was running before and off after, something stopped it while the logger was blind.



Orange rows: hot and running before the gap, fully off after it. Green rows: same state on both sides.

Why these are not trips

• **No limit was close.** When the two hot gaps began, the exhaust read 121-125 °C against a 200 °C trip limit, and the stack read 563-595 °C against 720 °C.

• **The "after" pattern is not the emergency-stop pattern.** A real emergency stop closes every gas valve but leaves one natural-gas vent valve open. After both hot gaps, every valve was closed, including that vent. That looks like the controls restarting into a safe "all off" state.

• **The logger broke first.** On 18 Feb the values freeze from about 17:48, then read -1 (the logger's "no signal" value). On 20 Mar the file changes its column layout three times (697, 577, 687 columns), the same mid-day format switch already flagged for that day.

• **The operator brought the plant straight back.** On 18 Feb air and heater were back within 6 minutes of the data returning. On 20 Mar the same heating step resumed at 12:56. Nobody started a cool-down, which is what follows a real emergency.

The three harmless gaps

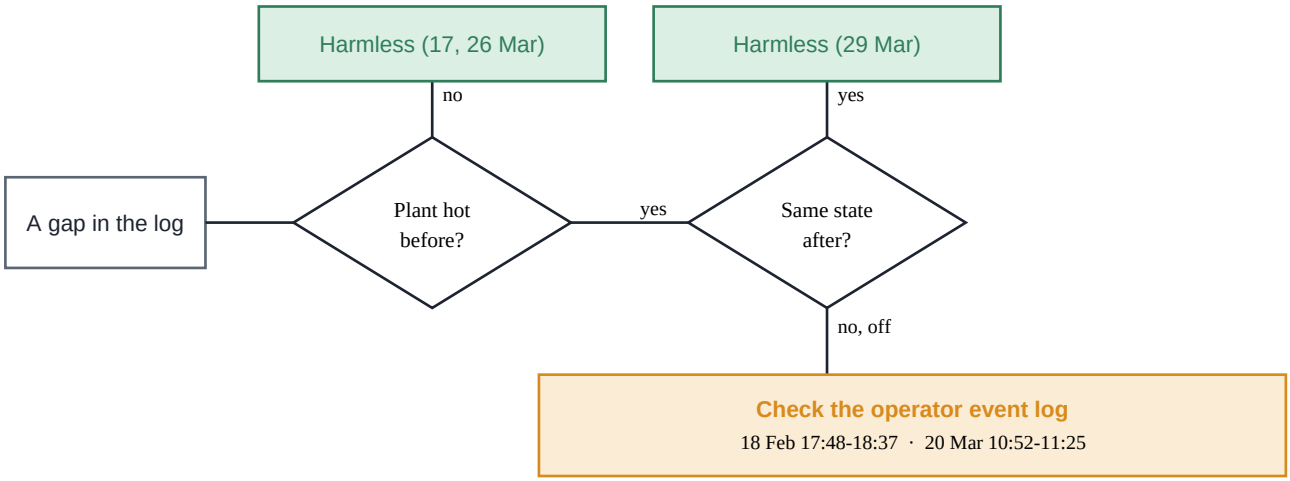
• **17 and 26 Mar:** the plant was cold (about 50-60 °C) with nothing on. On 17 Mar the sensors read -1 then 0, so the instruments were powered down, probably for maintenance. On 26 Mar the logger wrote no rows for 4.4 hours.

• **29 Mar, 02:00-03:00:** the switch to summer time. Clocks jumped from 02:00 to 03:00, so this hour never existed. The plant was heating at the same power on both sides.

If you remember one thing: only 18 Feb and 20 Mar need a second look, and the data cannot say what stopped the plant there.

Ask for two time windows from the event log

The logger does not record the safety controller's state or why it stopped. The plant's own alarm/event history does. Two windows are enough to close the question.



How each gap was sorted. Only the orange box needs someone with access to the plant's event history.

Questions to send to the plant operator

• **18 Feb 2026, 17:48-18:37:** was there a power cut, a controller restart or a communication loss? Did anyone press an emergency stop?

• **20 Mar 2026, 10:52-11:25:** was the control or logging software changed or restarted? The file layout changes at the same moment.

• **22 Mar 2026:** the day file exists but is empty (0 bytes). Was the logger off all day, or was the file lost?

Small fixes on our side

• Treat the 29 Mar 02:00-03:00 hour as a clock change, not as missing data.

• Treat values frozen for several minutes followed by -1 readings as a logger fault, as on 18 Feb.

Source: 1-second files in sofc-data/parquet, stage labels from operating_states.py, emergency events from trip_check.py.

If you remember one thing: send the operator two time windows, 18 Feb evening and 20 Mar late morning, and the question is closed.